

Single Goal Set: A New Paradigm for IT Megaproject Success

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Focusing IT project management on implementing a single overarching goal might substantially increase project successes.

In 1985, Alfred Spector and David Gifford interviewed a preeminent bridge designer, Gerard Fox, to compare bridge building with computer systems design. They wanted to find commonalities as well as differences that could help improve software development processes:

Civil engineering is one of the most classical of the engineering disciplines—it has well-developed designs, procedures, and tools at its disposal. Bridges rarely fall down. In contrast, computer systems design is one of the least classical of the engineering disciplines, and its products are often poorly understood, unmanageably complex, and unreliable.¹

Over the 20 years since, software engineering and project management have made giant strides in terms of concepts, methodologies, and automated tools. Yet the track record on software projects remains poor. In a 2001 update of its 1994 CHAOS report, the Standish Group did find some improvements in software development project success.^{2,3} However, of 280,000 projects surveyed, 137,000 were still late, over budget, or both; and 65,000 failed outright.

These failures aren't specific to any product and seem to be free of vendor and implementation bias as well. Describing massive distribution problems from a flawed implementation of SAP's R/3 enterprise resource planning (ERP) system, IT journalist Malcolm Wheatley reported in *CIO* magazine that the "blame" was evenly spread around: "SAP implementations are no more likely to go down the tubes than ERP systems from other ven-

dors," Wheatley wrote. "An alarmingly long list of top-drawer integrators have fallen flat on their faces."⁴

Large IT projects seem especially vulnerable. Whether they are ERP systems, customer relationship management (CRM) or supply chain management (SCM) systems, or any similarly complex IT initiative, the scenario of delays, dissatisfied clients, and horrendous cost overruns is common. "IT and business managers are famous for making commitments without enough knowledge or information and for setting unrealistic timetables for decisions and actions," reports Gartner Research, a technology market research firm.⁵

Not understanding the relationship boundaries between an IS organization and its significant partners ... contributes to unknown and unacceptable levels of performance. Many traditional IT projects fail the "on time, on

budget” goal because of internal politics, a mismatch between technologies and staff skills, and a constant tendency to rethink requirements.⁵

Whether or not we agree with Gartner’s diagnosis, we must concur with their observation that IT projects are notorious for failing to meet their objectives and leaving behind dissatisfied clients and bruised implementation teams.

Could our way of viewing IT megaproject goals be at the root of failed implementations? Could a different paradigm “cure” the problem?

A megaproject perspective

Climbing Mount Everest was also a megaproject. It involved a challenging goal, large teams, several intermediate milestones, and unpredictable extraneous circumstances. It experienced major overruns of money, time, even lives. Was the project a success? The answer is a categorical yes. I postulate that we think of it this way because the project had an overarching goal—reaching the peak—and succeeded in achieving it.

We could make the same case for landing men on the moon. These ultimate successes negated how much the cost overran the original budgets and how many intermediate milestones weren’t met. Achieving the single goal gave the project the label of success.

This does not mean that cost or time overruns are unimportant. It means only that once the project achieves its key goal, it is considered to be successful. Subordinate goals are relegated to subsequent projects, and the teams that implement them work from a cultural mindset based on success. This mindset motivates success in achieving the subordinate goals as well.

Managing user expectations

Focusing on a single goal also helps manage user expectations from the project. My experience observing several project implementations suggests that unmet expectations are a major cause of project failures. The original Chaos report confirms this observation.³ This report identified various project success factors. Those addressing aspects of user expectations included a “clear statement of requirements,” “clear vision and objectives,” “realistic expectations,” and “ownership” of the project. Together these factors accounted for nearly 30

percent of the positive indicators for project success. All these factors are related to managing user expectations, so any methodology that helps in this task offers a good starting point for developing a new IT project management approach. Taking a single or key goal approach is a step in this direction.

I’ve found that IT initiatives for which an organization has few expectations fare better in their implementations. For example, an organization that implements an ERP system because an aging legacy application will soon go out of support can easily define a clear and limited goal—namely, to implement a system that at least meets the minimum functionality of the existing legacy system. So long as this is the case, the implementation remains relatively straightforward and the project goes through smoothly.

IT methodology revisited

IT project management borrowed from traditional engineering disciplines to develop its own body of knowledge. The classic waterfall process model is a case in point. As a model of software project management, it derives from staged process charts common to other engineering projects. Most large IT project implementation methodologies, such as SAP’s Accelerated SAP model, are variations of this basic model.

However, building software differs from building a bridge. For one thing, software is *inherently*, not *accidentally*, complex, as Frederick Brooks describes it in “No Silver Bullet.”⁶ For example, it doesn’t scale up by a mere repetition of the same element in larger sizes: “In most cases, the elements interact with each other in some nonlinear fashion, and the complexity of the whole increases much more than linearly.”⁶

To this complexity, Brooks attributes “the difficulty of communications among team members, which leads to product flaws, cost overruns, schedule delays.” In IT projects such as ERP and SCM, the additional complexities of client teams and conflicting business imperatives sully the already murky waters of software development. Nor do the geopolitical complexities of this millennium’s global delivery model simplify the problem.

Agile or lean methods represent the latest development methodology for managing IT projects.⁷ The agile movement’s proponents

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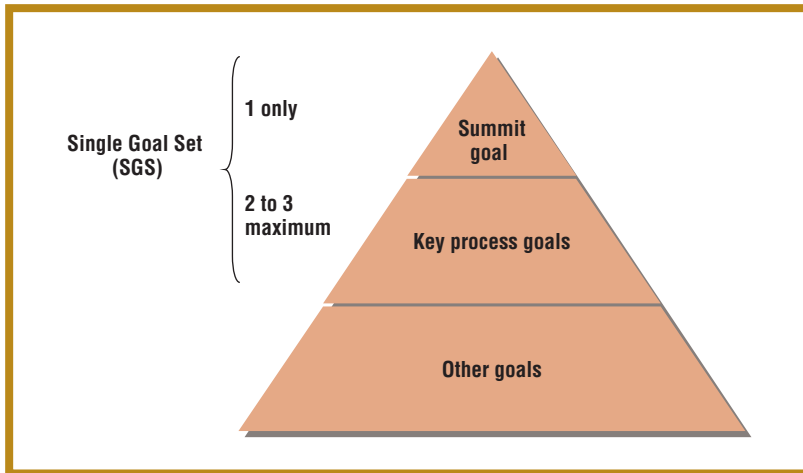


Figure 1. Single Goal Set. A summit goal reflects the project's overarching purpose. The methodology can support two or three subprocess goals, if necessary to achieve consensus on large, complex systems.

view organizations as complex adaptive systems in which requirements are emergent.⁸ Thus, they propose development methodologies that allow a project to emerge rather than follow a detailed plan. This gives rise to iterative approaches that respond to continuing feedback. Of course, agile methods are not an entirely new idea; spiral, incremental, and other such methodologies are variants of this theme.

The movement has fierce loyalists and equally vocal antagonists. One criticism is that reacting to every micro change event, rather than to major milestones, could compromise the eventual delivery. Recognizing this, Barry Boehm has recommended that large projects using agile methods establish at least three major milestones “to serve as project progress indicators.”⁹ In other words, preventing large projects from spiraling out of control requires some sort of centripetal force.

Toward a Single Goal Set paradigm

Combining this centripetal control force with an overarching project goal could give rise to a new methodology for successful project implementation. The approach would focus on a single project goal—one that all entities in an organization can appreciate and own.

Ideally, the methodology requires zeroing in on a single goal for the whole enterprise, but building consensus around a single goal for a large project might be impossible. Therefore, a project might have to refine the method by adding a selected few subprocess goals to the single overarching goal. In practice, therefore, an organization would deploy the methodology by agreeing first on a core process and a single goal for this process,

called the *summit goal*. Then the organization would agree on two or, at most, three additional vital processes and define goals for them. Ideally, it would define these process goals in terms of a specific transaction event, which is easy to track or measure, rather than a business deliverable, which is more amorphous and difficult to pin down. The summit goal and the two or three vital process goals become the *Single Goal Set*, as shown in figure 1, for the IT project implementation.

The SGS is the centering force for the project—like a gyroscope, always keeping the project from spiraling out of control. The project implementation is iterative. The first iteration aims solely at achieving the SGS. Subsequent iterations seek to achieve subordinate goals. The process therefore incorporates the flexibility of agile methods, while ensuring the overall goal-oriented discipline of traditional planning methodologies. It is agile in its simplicity and iterative nature. It is disciplined in its overarching implementation goal, understood and accepted by all.

The SGS methodology comprises 10 steps:

- Identify the organization's core process for the project.
- Identify any other key processes—the fewer the better, to a maximum of three.
- Define a single goal for the core process—that is, the summit goal—and goals for each key process. These become the SGS for the implementation program.
- Identify subordinate goals, although they apply to other processes and aren't the focus of the first implementation iteration.
- Work intensely to build consensus around the project's SGS.
- Use internal communication to stress the IT project's SGS.
- Focus the implementation team only on achieving the SGS. Subordinate goals are not addressed at all in the first iteration.
- When the project meets the SGS, declare the project successful.
- Identify gaps in the subordinate goals and assess their organizational impact.
- In subsequent iterations, bridge the gaps and achieve the subordinate goals.

This methodology intrinsically optimizes resource allocation. Time, money, effort, and other investments are spent first on realizing

Table 1**Example summit goal statements for major IT initiatives**

Type of company	Key megaproject driver	SGS statement
Large multilocation manufacturing company	■ Had to phase out legacy system because its relational DBMS was going out of support in one year ■ Wanted to consolidate all businesses/locations and close the books across the company	1. Consolidate companywide books accurately by second day of each month. 2. Have an alternate system in place when the legacy system goes out of support.
Manufacturing subsidiary of an international company	■ Had to integrate to the parent company ERP within six months	1. Integrate key business processes—general ledger and sales order processing, inventory—with parent company.
Discrete manufacturing unit	■ Had to effectively manage an inventory spread across a large number of subcontractors. Current system was ad hoc manual and legacy. Auditors had passed strictures at the last audit and the board had mandated that a system be put in place.	1. Connect to subcontractors and monitor stocks on a real-time basis.
Fast-moving consumer goods retailer	■ Was operating a countrywide distribution system with archaic reporting systems ■ Wanted to get a customer pulse that could help bring products to market quickly and take on very aggressive competition.	1. Connect to dealer network. 2. Capture customer feedback from specified <i>touch points</i>—that is, those interfaces where the customer interacts with the organization.

the organization's most important goals. Only after achieving the SGS does the project take up subordinate goals for implementation. Traditional methodologies often lack this sharp focus and result in a project that spends its budget with none of the goals fully satisfied. Any additional budget goes to rescuing what now looks like a sinking ship.

Can an entire organization adopt just one summit goal? I believe it can. In fact, I postulate that most organizations have one powerful driver or goal that leads them to attempt the IT initiative in the first place. Somewhere along the line, in the process of scoping the IT project, this goal gets diffused. Consultants and product vendors inundate the company with a large set of possible benefits that raise the organization's expectations to impossible levels. When these expectations fail to materialize, we are left with yet another failed project.

I did a quick survey of five diverse companies. All five had recently implemented large IT projects, and all were completely dissatisfied at the project's end. I asked them just one question: "Why did you go for this major IT initiative?" Table 1 summarizes their responses. It also suggests a summit goal or SGS that corresponds to their answers.

Table 1 shows that all five companies had one or two overriding reasons for venturing

on a large IT project. The core of the suggested methodology is to capture these reasons in a goal statement and keep it in the forefront of the project implementation as a summit goal or SGS.

Proposed methodology

The 10 SGS steps fit into a four-stage project management methodology that makes no major departure from classical implementation methodologies. The difference is only in keeping the project focused on a very small number of sharply defined, well-understood, and agreed-upon goals for the implementation.

Define and adopt stage

In this stage, the methodology asks key persons in the organization to answer the question, "What is the single most important benefit I wish to see at the end of this IT project?"

Of course, different role holders in the company would answer the question differently. The project manager or another designated *project champion* facilitates the process of arriving at a consensus—or near consensus—on the implementation SGS. The project champion must have sufficient organizational authority to lead this crucial phase of the project.

Once the project achieves consensus, the core implementation team starts the process of

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adopting the SGS across the organization. This is also largely a change management initiative and requires using all the standard organizational methods for bringing about positive change. The team publicizes the SGS as the IT project's purpose. The publicity must clarify the SGS benefits to the organization and to individual departments.

This stage ends when most people in the organization identify with and take ownership for achieving the SGS.

Create stage

In the second stage, the team creates the implementation blueprint, focusing only on achieving the SGS. The team maps any product functionality to fit the SGS deliverable. The methodology requires all workarounds, process changes, and customizations to aim at the single purpose of achieving the SGS, and it allows no compromise of this purpose.

Implement stage

The team then takes up the blueprint for implementation. This stage requires periodic evaluations to ensure that SGS deliverables fully satisfy all stakeholders. Where necessary, the team might have to integrate with legacy systems, create workarounds, and resort to additional data entry to ensure that the SGS deliverables are not compromised. Meeting the SGS successfully will greatly reduce the pressures on the organization and project team. The general bonhomie that follows success facilitates the subsequent iterations of the implementation to take care of suboptimizations that crop up in the single-minded pursuit of the SGS.

Evaluate and iterate stage

Finally, the project takes up subordinate goals for review and implementation after careful cost-benefit evaluation. Goals that don't justify the investment are dropped; other goals might be postponed, and the rest are taken up for implementation. Because the project has already achieved its principle objectives, business benefits have already started to accrue. Further investments in time, energy, and money only add to the benefits set.

Example project

Consider an organization that couldn't close its books immediately after the end of an

accounting period. As part of an international conglomerate, the organization receives a corporate mandate to finalize its accounts and post results by the first day of every quarter. To meet this mandate, the organization is contemplating a change to a new ERP system. In addition to the corporate mandate, the organization wants to create systems that give it an accurate view of channel inventory on a real-time basis—something current practices don't support.

The organization's SGS then consists of two goals:

- Accurately close the books and declare results by the first day of every quarter.
- View the channel inventory on a real-time basis.

The IT project implementation focuses only on these goals. It creates the systems, subsystems, and process changes required to achieve them in the implementation. Once the project achieves these goals to the satisfaction of all the process participants, the organization announces the project's successful completion.

Only then, during evaluation and iteration, does the project team evaluate other process goals. Let's say the evaluation finds that manual data entry has become necessary at two manufacturing locations. The project resolves this problem by creating a subsystem, process change, or workaround.

The methodology implements these subsequent steps "off line," as it were. That is to say, the problems are localized to a small group of users and don't color the success of the whole project. The project can tackle local issues and problems without pressure and noise that might jeopardize the overall project success.

Suboptimization issues

Critics of this methodology might say it opens the door to suboptimizations—for example, redundant data capture or omitted subprocesses in the first iteration—because it lacks a total process view up front. However, I call these *planned suboptimizations*—something akin to planned redundancy in a normalized table to achieve higher retrieval efficiency. The crux of the methodology is to meet the most important implementation drivers and not to compromise the business needs they represent.

The evaluation stage would consider all such suboptimizations and their impact on cost, quality, and deliverables. If justified from a cost-benefit perspective, the project would take them up for resolution through workarounds or customizations as appropriate.

Large IT projects lack a consistently satisfied client because they don't optimally allocate resources toward achieving the most vital organizational business goals. Inevitably, a resource crunch builds up by the time the project reaches midlife. A huge negative sentiment also accumulates, which further vitiates project success. To address this endemic problem, the SGS project methodology ensures that resources are first allocated to the critical deliverables. Achieving the SGS establishes a successful frame of mind for the project team, which also adds to the overall acceptance of the IT solution in the organization.

IT project management must find answers for the malaise of myriad failed projects. The situation calls for a new paradigm. Perhaps a simple, first-principles approach is the answer. ☞

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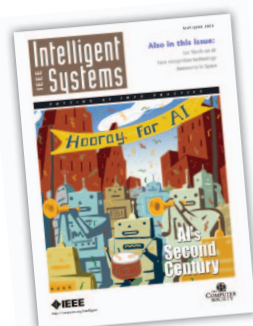


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